Isaac Oliva-González

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Summary

Researcher in computer science with a background in chemical engineering. My work focuses on mathematical optimization and operations research, machine learning, and quantum and hybrid quantum-classical methods. I develop and analyze mathematical programming models for large-scale, structured decision problems, with emphasis on exploiting problem structure to obtain stronger formulations and tractable solution methods; applications include scheduling, routing, production planning, and process systems.

Research Interests

Optimization & Operations Research

Linear, mixed-integer, and network optimization; algebraic and combinatorial structure

Quantum Optimization & Algorithms

Hybrid Quantum-Classical Methods, Quantum Machine Learning, Quantum Annealing

Applications

Energy and process systems; scheduling, routing, production and capacity planning

Education

MSc in Computer Science *Aug 2023 - Aug 2025*

Universidad Autónoma de Querétaro Querétaro, MEX

- Dissertation on quantum annealing and simulated annealing for discrete-space route optimization.

BSc in Chemical Engineering *Jan 2017 - Dec 2021*

Universidad de Guanajuato Guanajuato, MEX

- Dissertation on modeling and design of intensified distillation sequences for renewable fuel separation.

Experience

Tecnológico de Monterrey *Jan 2025 - Aug 2025*

Visiting Research Scholar (Remote) | *Supervisor: Dr. Ilse María Hernández-Romero* Monterrey, MEX

- Conducted collaborative research on hybrid quantum–classical neural architectures using variational quantum circuits to model black carbon dynamics, benchmarking against classical models to assess the contribution of quantum-enhanced architectures in predictive performance and model expressiveness (*PennyLane*). See [J1].

Universidad Autónoma de Querétaro *Aug 2023 - Aug 2025*

MSc Student Querétaro, MEX

- Designed and benchmarked quantum and classical optimization algorithms for discrete shortest-path problems, analyzing performance trade-offs across solution quality and computational efficiency (*Julia*). See [J2].
- Designed and implemented finite automata for encoding alphabetic sequences and performing operations on deterministic finite automata (*Mathematica*).

Ingredion México *Feb 2022 - Dec 2022*

Process Technician San Juan del Río, MEX

- Operated and monitored industrial starch processing systems, performing on-site troubleshooting and real-time adjustments to ensure process reliability.

NOVARES México *Jan 2021 - Jun 2021*

Undergraduate Intern Silao, MEX

- Supported the implementation of health, safety, and environmental (HSE) programs in an automotive manufacturing environment.

Instituto Tecnológico de Villahermosa *Summer 2019*

Visiting Research Scholar | *Supervisor: Dr. Francisco López-Villarreal* Villahermosa, MEX

- Developed a computational interface for vapor–liquid equilibrium calculations in isothermal flash distillation, applying numerical methods to thermodynamic optimization problems (*MATLAB*).

Research Projects

Exploratory Study of Symmetry in Combinatorial Optimization *Julia, JuMP*
Available here 2026

- Exploratory study of symmetry reduction in combinatorial optimization (ILP), including shortest path, graph coloring, max-cut, and TSP. Applied lexicographic constraints, variable fixing, and automorphism-based methods to reduce redundant solutions.

Symmetry-Aware CC Unit Commitment *Julia, JuMP*
Available here 2026

- Study of symmetry reduction applied to the *Combined Cycle Min-Up/Min-Down Unit Commitment Problem* (CC-MUCP). Derived demand-aware lexicographic constraints to break the two-level permutation symmetry from identical packages and gas turbines.

SDP-Symresack: Symmetry Handling in SDP
Available here 2026

- Explored whether lexicographic ordering from symresacks can be adapted from binary integer programming to semidefinite programming, using MaxCut on K_3 as a worked example.

Corrugator Production Scheduling *Julia, JuMP*
Available here 2026

- Mixed-integer programming model for corrugator production scheduling. Groups compatible orders into cutting patterns over discrete roll widths, minimizing setups subject to a trim-waste cap and a line-balance constraint.

Skills

Programming: Python, Julia
Optimization & Modeling: JuMP, GAMS, MATLAB, OR-Tools (Gurobi/CPLEX/HiGHS)
Quantum & Scientific Computing: PennyLane, Qiskit, Mathematica, Aspen Plus, Jupyter Notebooks

Languages

Spanish: Native proficiency.
English: Proficient (TOEFL ITP 562 | CEFR B2).
German: Elementary proficiency (CEFR A2).

Research & Publications

Journal Articles

[J1] **A Hybrid Quantum-Classical Machine Learning Framework for Black Carbon Forecasting** *EPJ Quantum Technol.*
I. Oliva-González, L. T. González, O. D. Lara-Montaño, A. I. Ramirez, A. Mendoza, I. M. Hernández-Romero 2026
EPJ Quantum Technology, DOI: 10.1140/epjqt/s40507-026-00511-0

[J2] **A QUBO-Driven Simulated Annealing Methodology for Solving the Shortest Path Problem in Urban Transportation Networks** *Algorithms*
I. Oliva-González and H. Jiménez-Hernández 2026
Algorithms, DOI: 10.3390/a19050352

Book Chapters

[B1] **Integrating Quantum Computing into Sustainable Carbon-Capture Materials Research: Opportunities and Perspectives** *Hand. Carbon Neg. Footpr. Mater.*
I. M. Hernández-Romero, **I. Oliva-González**, O. D. Lara-Montaño, K. Hernández-Romero, G. G. Esquivel-Patiño, O. Kharissova, A. I. Ramirez, L. T. González 2026
Handbook of Carbon Negative Footprint Materials, Springer, Cham, DOI: 10.1007/978-3-031-87501-4_60-1

Conference Proceedings

[C1] **Síntesis de columnas con múltiples paredes divisorias para la separación de combustible sustentable de aviación** — *Av. Ing. Quím.* 2024

I. Oliva-González, A. G. Romero-Izquierdo, C. Gutiérrez-Antonio, F. I. Gómez-Castro, and S. Hernández

Avances en Ingeniería Química, vol. 3, no. 1, pp. 265–270, AMIDIQ 2024. Available online

[C2] **Computer-aided design of intensified separation sequences for a complex mixture of renewable hydrocarbons** — *Comput. Aided Chem. Eng.* 2024

I. Oliva-González, A. G. Romero-Izquierdo, C. Gutiérrez-Antonio, F. I. Gómez-Castro, and S. Hernández

ESCAPE34/PSE2024, Elsevier, vol. 53, pp. 1483–1488, DOI: 10.1016/B978-0-443-28824-1.50248-9

[C3] **Síntesis de columnas con múltiples paredes divisorias para la separación de combustible sustentable de aviación** — *Av. Ing. Quím.* 2023

I. Oliva-González, C. Gutiérrez-Antonio, F. I. Gómez-Castro, S. Hernández, and A. G. Romero-Izquierdo

Avances en Ingeniería Química, vol. 2, no. 1, pp. 402–407, AMIDIQ 2023. Available online

[C4] **Síntesis de secuencias intensificadas de destilación para la separación de una mezcla multicomponente de hidrocarburos renovables** — *Av. Ing. Quím.* 2022

I. Oliva-González, C. Gutiérrez-Antonio, F. I. Gómez-Castro, E. Quiroz Pérez, S. Hernández, and A. G. Romero-Izquierdo

Avances en Ingeniería Química, vol. 1, no. 4, pp. 218–223, AMIDIQ 2022. Available online

Conference Presentations

[P1] **Síntesis de columnas con múltiples paredes divisorias para la separación de combustible sustentable de aviación** — 2023

I. Oliva-González, A. G. Romero-Izquierdo, C. Gutiérrez-Antonio, F. I. Gómez-Castro, and S. Hernández

Retos y Oportunidades al Net-Zero 2030, Querétaro, México (Oral presentation)

[P2] **Synthesis of alternative intensified sequences for the separation of a multicomponent renewable hydrocarbons mixture** — 2022

I. Oliva-González, A. G. Romero-Izquierdo, C. Gutiérrez-Antonio, F. I. Gómez-Castro, and S. Hernández

12th International Conference Distillation & Absorption 2022 (DA2022), Toulouse, France (Poster presentation)